Appendix D: Market structure and concentration methodology and UK share of supply by revenue

Introduction

D.1 This appendix provides additional detail on the methodology for calculating shares of supply as set out in chapter 3 and UK shares of supply by revenue. We present:

(a) shares of supply based on UK revenues and overall revenue growth based on UK revenues.

(b) the methodology used to calculate shares of supply by revenue, including;

(i) the categorisation of provider revenues and limitations of provider data;

(ii) the calculation of the market size; and

(iii) notes on the HHI in chapter 3.

(c) notes about the data used to calculate:

(i) shares of supply by capacity; and

(ii) shares of supply by flows of new business.

UK shares of supply by revenue

D.2 In the following section we present shares of supply for IaaS, PaaS and IaaS with PaaS. As outlined in chapter 3, we have considered where available shares which align with our findings on the geographic market for the cloud services markets, which is UK+EEA. However, UK only data is also informative of providers' success in competing for UK customers and therefore useful in assessing their strength in the markets concerned.

D.3 Shares of supply based on UK revenues are consistent with those based on the UK+EEA geographic area. They suggest that UK cloud services markets are slightly more concentrated than at the UK+EEA level.

UK shares of supply for IaaS

D.4 UK shares of supply for IaaS for 2020 to 2024 are presented in the table below. These shares include revenues from IaaS based on accelerated compute which

could not be distinguished across all Parties' data, but generally we understand the contribution of accelerated compute to the total revenues to be small.

Table D.1: UK shares of supply for IaaS, 2020 – 2024

%

	2020	*2021*	*2022*	*2023*	*2024*
AWS	40-50]%	40-50]%	40-50]%	[40-50]%	[40-50]%
Microsoft	[30-40]%	30-40]%	30-40]%	[30-40]%	[30-40]%
Google	[5-10]%	[5-10]%	[5-10]%	[5-10]%	[5-10]%
IBM	[0-5]%	[0-5]%	[0-5]%	[0-5]%	[0-5]%
Oracle	[0-5]%	[0-5]%	[0-5]%	[0-5]%	[0-5]%
Other	[10-20]%	10-20]%	[10-20]%	[10-20]%	[10-20]%

Source: CMA analysis of first-party revenue data and IDC and Synergy data. Shares may not sum to 100 due to rounding. 'Other' category includes the other firms included in the IDC and Synergy data sets (see chapter 3, paragraph 3.180).

D.5 Our analysis also shows that:

(a) AWS is the largest provider of IaaS in the UK, and between 2020 and 2024, its share has remained broadly stable between [40-50]% and [40-50]%;

(b) Microsoft is the second largest provider of IaaS in the UK. Its share has increased from [30-40]% in 2020 to [30-40]% in 2024;

(c) Google is the third largest IaaS provider and has a significantly lower share of IaaS revenues than AWS and Microsoft: it's share has grown from [5-10]% in 2020 to [5-10]% in 2024.

(d) For IBM and Oracle, shares have remained in the [0-5]% range from 2020 to 2024.[1]

(e) The combined share of all other IaaS suppliers has decreased from [10-20]% in 2020 to [10-20]% in 2024.

D.6 The evidence shows that the IaaS market in the UK is highly concentrated.

(a) AWS' and Microsoft's combined share of supply for IaaS was increasing between 2020 and 2022, and remained high in 2023 and 2024 at [70-90]% and [70-90]% respectively. Our analysis shows that concentration in IaaS is stable in the sense that the combined share of AWS and Microsoft has remained persistently high from 2020 – 2024 [70-90]%.

[1] CMA analysis of first-party revenue data and IDC and Synergy data. Responses to the CMA's information requests [✂].

(b) The next largest provider is Google, but in 2024 AWS remained around five times its size. Google is growing and gaining share, but the combined market share of Microsoft and AWS has continued to grow since 2020.

UK shares of supply for PaaS

D.7 UK shares of supply for PaaS for 2020 – 2024 are presented in the table below.

Table D.2: UK shares of supply for PaaS, 2020 – 2024

%

	2020	*2021*	*2022*	*2023*	*2024*
AWS	[20-30]%	[20-30]%	[20-30]%	[20-30]%	[20-30]%
Microsoft	[20-30]%	[20-30]%	[20-30]%	[20-30]%	[20-30]%
Google	[5-10]%	[5-10]%	[5-10]%	[10-20]%	[5-10]%
IBM	[0-5]%	[0-5]%	[0-5]%	[0-5]%	[0-5]%
Oracle	[0-5]%	[0-5]%	[0-5]%	[0-5]%	[0-5]%
Other	[30-40]%	[30-40]%	[30-40]%	[30-40]%	[30-40]%

Source: CMA analysis of first-party revenue data and IDC and Synergy data. Shares may not sum to 100 due to rounding. 'Other' category includes the other firms included in the IDC and Synergy data sets (see chapter 3, paragraph 3.180).

D.8 The PaaS market is less concentrated than IaaS. Our analysis shows that:

(a) AWS is the largest provider of PaaS. Its share decreased from [20-30]% in 2020 to [20-30]% in 2024;

(b) Microsoft is the second largest provider of PaaS and its share has increased from [20-30]% in 2020 to [20-30] in 2024;

(c) Google is much smaller than AWS and Microsoft and its share has increased from [5-10]% in 2020 to [10-20]% in 2023, decreasing slightly to [5-10]% in 2024;

(d) For IBM and Oracle, shares, have remained in the 0-5% range from 2020 to 2024.[2]

D.9 The combined share of the other smaller cloud providers and ISVs has remained stable at [30-40]% in 2020 to [30-40]% in 2024.

D.10 Our analysis also shows that concentration in PaaS is stable with the combined share of AWS and Microsoft remaining between [40-60]% and [40-60]% between 2020 and 2024.

[2] CMA analysis of first-party revenue data and IDC and Synergy data. Responses to the CMA's information requests [✂].

UK shares of supply for IaaS and PaaS combined

D.11 UK shares of supply for IaaS and PaaS combined for 2020 – 2024 are presented in the table below.[3]

Table D.3: UK shares of supply for IaaS and PaaS combined, 2020 – 2024

%

	2020	*2021*	*2022*	*2023*	*2024*
AWS	[30-40]%	[30-40]%	[30-40]%	[30-40]%	[30-40]%
Microsoft	[20-30]%	[30-40]%	[30-40]%	[30-40]%	[30-40]%
Google	[5-10]%	[5-10]%	[5-10]%	[5-10]%	[5-10]%
IBM	[0-5]%	[0-5]%	[0-5]%	[0-5]%	[0-5]%
Oracle	[0-5]%	[0-5]%	[0-5]%	[0-5]%	[0-5]%
Other	[20-30]%	[20-30]%	[20-30]%	[20-30]%	[20-30]%

Source: CMA analysis of first-party revenue data and IDC and Synergy data. Shares may not sum to 100 due to rounding. 'Other' category includes the other firms included in the IDC and Synergy data sets (see chapter 3, paragraph 3.180).

D.12 Our analysis shows the shares of supply in cloud services by revenue – that is IaaS and PaaS in combination. Our analysis shows that:

(a) AWS is the largest provider of IaaS and PaaS combined in the UK and its share has decreased from [30-40]% in 2020, to [30-40]% in 2024;

(b) Microsoft is the second largest provider of IaaS and PaaS combined and its share has increased from [20-30]% in 2020 to [30-40]% in 2024;

(c) Google is the third largest provider of IaaS and PaaS combined and has a significantly lower share than AWS and Microsoft: Google is significantly smaller than AWS and Microsoft, and in 2024 has reported less than a quarter of AWS' revenue. It has increased its share from [5-10]% in 2020, to [5-10]% in 2024.

(d) for each of IBM and Oracle, shares have remained in the 0-5% range from 2020 to 2024.[4]

D.13 The analysis also shows that concentration in IaaS and PaaS combined is largely stable over time, with the combined share of AWS and Microsoft remaining between [50-70]% and [60-80]% between 2020 and 2024.

[3] Similar to shares of supply for IaaS, these shares include revenues from IaaS based on accelerated compute which could not be distinguished across all Parties' data, but we understand the contribution of accelerated compute to the total revenues to be small.
[4] CMA analysis of first-party revenue data and IDC and Synergy data. Responses to the CMA's information requests [✂].

D.14 The combined share of the other smaller cloud providers and ISVs has remained largely stable, between [20-30]% and [20-30]% between 2020 and 2024.

Shares of year-on-year UK IaaS revenue growth

D.15 For each firm we calculated its annual growth in IaaS revenues, then divided this by the sum of annual revenue growth across all providers to give the share.[5]

D.16 UK shares of year-on-year IaaS revenue growth for 2021 to 2024 are presented in the table below.

Table D.4: UK shares of year-on-year IaaS revenue growth, 2021 – 2024

				%
	2021	*2022*	*2023*	*2024*
AWS	[40-50]%	[40-50]%	[30-40]%	[30-40]%
Microsoft	[30-40]%	[40-50]%	[30-40]%	[30-40]%
Google	[5-10]%	[5-10]%	[10-20]%	[10-20]%
IBM	[0-5]%	[✂]	[0-5]%	[✂]
Oracle	[0-5]%	[0-5]%	[0-5]%	[0-5]%
Other	[0-5]%	[5-10]%	[5-10]%	[10-20]%

Source: CMA analysis of data from AWS, Microsoft, Google, IBM, Oracle and some smaller IaaS providers. Shares may not sum to 100 due to rounding. 'Other' category includes the other firms included in the IDC and Synergy data sets.

D.17 Shares of supply by year-on-year revenue growth in IaaS in the UK show that AWS and Microsoft have consistently held the largest shares of overall revenue growth. Google has the third largest share of overall revenue growth, consistent with its position in the market. [6]

Shares of year-on-year UK IaaS and PaaS revenue growth

D.18 For each firm we calculated its annual growth in IaaS plus PaaS revenues, then divided this by the sum of annual revenue growth across all providers to give the share.

D.19 UK shares of year-on-year IaaS and PaaS revenue growth for 2021–2024 are presented in the table below.

[5] Similar to the shares of supply for IaaS, the underlying data includes revenues from IaaS based on accelerated compute which could not be distinguished across all Parties' data, but we understand the contribution of accelerated compute to the total revenues to be small.
[6] CMA analysis of first-party revenue data and IDC and Synergy data. Responses to the CMA's information requests issued under section 174 to [✂].

Table D.5: UK shares of year-on-year IaaS and PaaS revenue growth, 2021 – 2024

%

	2021	*2022*	*2023*	*2024*
AWS	[30-40]%	[30-40]%	[30-40]%	[20-30]%
Microsoft	[30-40]%	[30-40]%	[20-30]%	[30-40]%
Google	[10-20]%	[5-10]%	[10-20]%	[5-10]%
IBM	[0-5]%	[✂]	[0-5]%	[✂]
Oracle	[0-5]%	[0-5]%	[0-5]%	[0-5]%
Other	[10-20]%	[10-20]%	[20-30]%	[20-30]%

Source: CMA analysis of data from AWS, Microsoft, Google, IBM, Oracle and some smaller IaaS providers. Shares may not sum to 100 due to rounding. 'Other' category includes the other firms included in the IDC and Synergy data sets.

D.20 Shares of year-on-year revenue growth in IaaS and PaaS combined in the UK show that AWS and Microsoft have consistently held the largest shares of overall revenue growth. Google has the third largest share of overall revenue growth, consistent with its position in the market.[7] Other providers have a larger share of revenue growth in IaaS and PaaS combined than the IaaS market, and this has been growing over the period 2021-2024.

Methodology for shares of supply by revenue

Categorisation of provider revenues and limitations of provider data

D.21 We asked providers to supply revenues by individual cloud services or service categories. We then mapped these into IaaS, PaaS, and SaaS using service provider mapping lists and additional descriptions submitted by the providers and calculated totals for IaaS, PaaS, and for IaaS and PaaS combined.[8] We reviewed the mapping of cloud services into IaaS, PaaS and SaaS used by Ofcom and adopted this in order to map individual cloud services or service categories into IaaS, PaaS and SaaS.[9]

D.22 In relation to the data provided directly to us by providers, we note that:

(a) One provider said it records relevant data in several different systems, some of which may not readily reconcile with each other. It therefore made certain adjustments to reconcile data across these systems.[10] We have adjusted the IaaS and PaaS totals based on the individual revenues by cloud service from one system such that the cloud provider's combined total matches the

[7] CMA analysis of first-party revenue data and IDC and Synergy data. Responses to the CMA's information requests [✂].
[8] We converted AWS, Microsoft, Google, IBM, [✂], Upcloud and Wasabi revenue data from USD to GBP using the Bank of England's average annual exchange rates for each year respectively. [✂] submission and some Upcloud data was submitted in Euros. For this we used the corresponding EUR to GBP exchange rate.
[9] We made one amendment to this mapping of cloud services. This amendment was to re-assign the category of cloud services labelled 'Management and governance' from SaaS to PaaS.
[10] [✂] response to the CMA's information request [✂].

aggregate revenues from another system. This methodology had the effect of increasing the cloud provider's shares by less than [✂] in each category in the UK and less than [✂] in the UK+EEA in IaaS and PaaS relative to just using revenues based on individual cloud services. This provider submitted that it has undergone a product mapping change based on its financial system which affected its 2024 revenue submission. [11] It provided details of this product mapping to its previous submission, and we used the previous submission to reconcile the service revenues and allocate these to IaaS, PaaS and SaaS based on the methodology outlined above.

(b) Oracle said that it was unable to provide annual revenue data based on calendar years and therefore it provided data according to its fiscal year, which ends on 31 May.[12] This means that Oracle's shares in any given year could be slightly overestimated or underestimated.

(c) IBM and Oracle did not provide revenue data segmented by individual cloud services. Oracle provided revenues grouped into service categories rather than segmented by individual cloud services – these service categories appear to follow broadly the mapping into IaaS and PaaS that we applied elsewhere.[13] [✂].[14] Given the shares of Oracle and IBM set out in chapter 3, any differences in the categorisation of their cloud services are unlikely to have a material effect on the final shares.

(d) We did not have access to the same level of information about each individual cloud service for some of the small UK IaaS providers (UpCloud, Civo, [✂], Hyve, Wasabi, [✂], Coreweave and Centerprise) as we did for AWS, Google and Microsoft. Therefore, for some providers we conducted desktop research to determine each cloud service's purpose and accordingly which subcategory it should fall into. We then applied the same mapping into IaaS, PaaS and SaaS as outlined above. As the small UK IaaS providers cumulatively totalled <1% of IaaS and PaaS combined revenues in the UK and UK+EEA regions in 2024, any differences in categorisation into subcategories, and therefore IaaS and PaaS based on misinterpretation of cloud services' purpose will not have a material effect on the final shares.

(e) [✂].[15] The Synergy data set contained revenues for CoreWeave, split into IaaS and PaaS. As we consider first party revenues to provide the best estimate of provider revenues, we allocated CoreWeave's first party revenues to IaaS and PaaS based on the proportion of IaaS and PaaS revenues reported in the Synergy data set for CoreWeave.

(f) Providers included revenues from IaaS based on accelerated compute which could not be distinguished across all Parties' data, but we understand the contribution of accelerated compute to the total revenues to be small.

[11] [✂] response to the CMA's information request [✂].
[12] Oracle's response to the CMA's information request [✂].
[13] Oracle's response to the CMA's information request [✂].
[14] [✂] response to the CMA's information request [✂].
[15] [✂] response to the CMA's information request [✂].

Therefore, the overall shares of supply will tend to largely proxy shares in IaaS based on standard compute.

D.23 Across several providers, the revenue data did not align with our definition of UK or EEA revenues, ie revenues generated from customers that are operating or trading in the UK or EEA. Although providers have identified UK or EEA customers on a different basis, we do not consider this to materially impact our results.

(a) One provider provided UK and EEA revenues by identifying revenues based on activity recorded to [✂] as a result of the customer's tax address, billing address, or customer address associated with the Account ID.[16]

(b) One provider identified UK and EEA revenue as revenues that were billed to the customers' billing address(es) in GB and EEA countries.[17]

(c) Three providers identified UK and EEA revenues as generated by customers with billing addresses in the UK.[18]

Calculation of the market size

D.24 In order to calculate the market size for IaaS and PaaS, we procured data from IDC and Synergy. We chose to procure data as it would not be practical to send statutory information requests to all companies providing cloud services in the UK. We procured data from two sources which allowed us to compare across data sets, and (if necessary) combine the reported figures based on our understanding of the underlying methodologies. We considered that using a combination of the two data sets would be the closest estimate of market size which matched our definitions of IaaS and PaaS. We chose IDC and Synergy as these were the only providers which were able to provide the data to our specification. Additionally, the selection of these two providers is consistent with the methodology Ofcom used to calculate shares by revenue for cloud services.[19]

D.25 Synergy provided data on revenues from 66 providers for IaaS across the UK and EEA regions and 39 providers for PaaS across the same regions. Synergy noted that relative to the data provided at PDR stage, it has made some amendments to the categorisation of revenue. These changes include re-categorisation of Content Delivery Networks as a cloud service, splitting out the GPUaaS and Generative AI-PaaS subcategories within IaaS and PaaS and changes to allocation of revenues following a re-organisation of some providers' services from IaaS and PaaS into SaaS. These changes increased Synergy's estimate of the IaaS market and decreased its estimate of the PaaS market in the UK in 2023 relative to the figures used in the PDR.[20]

D.26 IDC provided data on revenues from 41 providers for IaaS across the UK and EEA, and 296 providers for PaaS across the same regions. IDC noted that relative

[16] [✂] responses to the CMA's information requests [✂].
[17] [✂] responses to the CMA's information requests [✂].
[18] Responses to the CMA's information requests [✂].
[19] Ofcom cloud services market study, Annex A1. Share of Supply for cloud infrastructure services in the UK.
[20] Synergy [✂].

to the data used in the PDR, it has revised its model due to published financial insights and product mapping / estimation changes for several firms.[21] These changes have had the effect of increasing the market size for IaaS and increasing the market size for PaaS in the UK in 2023 relative to the figures used in the PDR.

D.27 We combined data from the two sources using a different methodology to calculate the market size for IaaS and PaaS respectively:

(a) IaaS: The two data sources combined reported revenues for 71 firms in the UK and 85 firms in the EEA. We considered that a number of these (across both data sets) did not currently supply services which aligned with our definition of IaaS as set out in chapter 3, therefore we excluded revenue for some firms at this stage. Where we did not have first party revenue, we then took an average of the revenue for each firm that appeared in both data sets. Where we did not have first party revenue, if a firm was present in only one data set, we included the revenue value as reported. After these steps, in order to calculate the total market size, we aggregated across the total revenues including first party revenues.

(b) PaaS: The two data sources combined reported revenue for 311 providers in the UK and 315 providers in the EEA. We did not consider it practical to assess whether each individual firm provided services that aligned with our definition of PaaS. IDC had the most comprehensive list of firms, however we considered some firms included to provide SaaS. Synergy had a shorter list of firms, however, it did not include some IDC firms which we considered provided PaaS. Therefore, after having replaced first-party revenues where available, we took the average of the revenue for each firm that appeared in both data sets. Then, within each data set, we aggregated the non-first party revenues which did not appear in both data sets and took an average across both data sets 'other' category to calculate an estimate of the rest of the market.

Herfindahl-Hirschman Index

D.28 The Herfindahl-Hirschman Index (HHI) can reflect both the number of firms in the industry and their relative size. It is defined as the sum of the squares of all the market shares in the market, and thus gives proportionately greater weight to the larger market shares.[22]

D.29 AWS and CMA estimates of HHI are presented in the Market structure and concentration section of chapter 3. Differences between these analysis and relevant caveats are explained here.

(a) AWS' reports HHI for IaaS and PaaS combined.[23] We calculated the HHI for IaaS, PaaS and IaaS and PaaS combined using the results of the shares of

[21] IDC [✂].
[22] CC3 (Revised), Guidelines for market investigations: Their role, procedures, assessment and remedies,, Annex A, paragraph 7.
[23] The Competitive landscape working paper, paragraph 1.3.

supply by revenue analysis. We note that we have defined separate markets for IaaS and PaaS, therefore a HHI calculated across IaaS and PaaS combined, as in AWS' analysis, does not accurately reflect the boundaries of the relevant product markets.

(b) Our HHI estimates and those submitted by AWS will be an overestimate. This is due to a number of firms being grouped into an 'other' category and the 'other' category being treated as a single firm in the HHI calculations. The overestimate will be greater for PaaS, compared to IaaS as the 'other 'category is larger for PaaS. In addition, this effect in PaaS is larger in our estimates of HHI than AWS' estimates, as our 'other' category was larger due to the method by which we combined the third-party data sources.

(c) AWS' analysis used third-party data from a research firm. Our analysis is also based on third-party research, but we have supplemented this with first party revenue data from 11 UK cloud providers which is likely to be more accurate. Further, AWS' analysis includes more providers for both IaaS and PaaS than our own analysis, which we refined to more closely match the relevant product markets.

Shares of supply by capacity

D.30 We gathered data from AWS, Google, Microsoft, IBM, Oracle, CoreWeave and some smaller IaaS providers that serve UK customers[24] on their data centre capacity in megawatts (MW) within UK+EEA,[25] globally, and in the UK, for the period 2020 to 2028[26]. In relation to this data, we note the following:

(a) In April 2025, Microsoft announced plans to increase its data centre capacity over the next two years.[27] We have not included this capacity as part of the forecasted figures presented in this analysis, as Microsoft submitted that unlike the data provided in response to our information request, this announcement is [✂].[28]

(b) Google provided realised (historic) data centre capacity for all regions up until the end of 2024.[29] Google provided [✂].[30]

[24] Centreprise, [✂], Scaleway and Wasabi.
[25] For the purpose of this analysis, we have allocated providers' capacities to UK+EEA if they were classified by the provider as relating to Europe.
[26] In the PDR we had data available only up until 2026. We requested updated data from providers with forecasts up until 2028, as well as providing the chance for providers to submit any updates to realised or forecast capacity. In response, some providers provided revised forecasts or historical capacity, therefore the shares presented may vary from those published at PDR.
[27] Microsoft announces new European digital commitments - Microsoft On the Issues, accessed 20 May 2025.
[28] Microsoft's response to the CMA's information request [✂].
[29] We note that [✂].
[30] Google's response to the CMA's information request [✂].

(c) Oracle provided data on its data centre capacity in the UK and EEA going back to 2021.[31] To estimate Oracle's UK and EEA capacity in 2020, we included an approximation of capacity based on Oracle's growth rate from 2020 to 2021 being in line with its average growth rate implied by its capacity data and projections for 2021 to 2026. We did this by calculating the average year-on-year capacity growth rate over the years 2021-2026 in the UK and EEA respectively and divided the 2021 capacity figures by these average growth rates. We also included this approximated UK and EEA capacity in 2020 towards Oracle's global capacity for 2020.

(d) Centerprise provided [✂].[32] [✂] provided [✂].[33] Scaleway provided [✂].[34] Wasabi provided [✂].[35] For these providers, we have not approximated their historic capacity.

(e) Some smaller cloud providers did not submit data on forecasted capacity for 2025 and 2026, specifically; [✂] did not provide forecasts for 2025 and 2026,[36] and [✂] did not provide forecasts for 2026.[37] In these cases, we have approximated the forecast capacity for each region to be equal to the final year for which we have capacity data (actual or forecasted) for that provider, as if the provider forecasts to maintain all of its existing capacity in each region, and not expand further (or reduce capacity). These cloud providers both expanded between 2023 and 2024; if they continue to expand their capacity to 2026, then the future shares for 'Other' will be higher than the shares we present in those years.

(f) Some smaller cloud providers ([✂][✂]) did not submit capacity data in comparable units (MW) in time, and as such, we have excluded these providers from our analysis.[38] We do not expect this to affect the findings in a meaningful way, as we are not aware of these providers having capacity of a scale that would affect our findings meaningfully.

(g) We received submissions on data centre capacity that included capacity from IaaS based on accelerated compute, which could not be distinguished in the Parties' data. We generally understand the contribution of accelerated compute to overall data centre capacity to be small.

Shares of supply by flows of new business

D.31 We gathered data from cloud providers on numbers of new customers, and the revenues associated with these new customers. Although each provider submitted data on a slightly different basis, we do not consider that this had a material effect on the analysis.

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[31] Oracle response to the CMA's information request [✂].
[32] Centerprise's response to the CMA's information request [✂].
[33] [✂] response to the CMA's information request [✂].
[34] Scaleway's response to the CMA's information request [✂].
[35] [✂] response to the CMA's information request [✂].
[36] [✂] response to the CMA's information request [✂].
[37] [✂] response to the CMA's information request [✂].
[38] Following the PDR, [✂] submitted that [✂]. [✂] response to the CMA's information request [✂]; [✂] response to the CMA's information request [✂].

(a) AWS provided data based on defining a newly acquired customer as one that spent at least $100 with it in a year, or alternatively for the purpose of specific questions, an acquired customer is defined as one that spent at least $100 with AWS in a year but nothing prior to that.[39]

(b) Microsoft provided data based on defining a newly acquired customer as one that spent at least $100 on its cloud services ('Azure Consumption Revenue') for the first time in a year.[40]

(c) Google provided data based on defining a newly acquired customer as one that spent at least $100 annual recurring revenue on its cloud services for the first time in a financial quarter starting from 2021.[41]

(d) Oracle provided data based on defining a newly acquired customer as one that appeared in its revenue data for the first time in a year for 2021–2023.[42] We then requested that Oracle provide data for 2024 based on defining a newly acquired customer as one who spent more than $100 for the first time in a year (in its revenue data).[43]

(e) IBM provided data for 2021–2023 based on defining a newly acquired customer as [✂].[44] IBM then provided data for 2024 based on defining a newly acquired customer as one who spent more than $100 for the first time in a year (in its revenue data).[45]

(f) For other providers,[46] we asked them to define a new customer as one who spent more than $100 for the first time in a year (in that provider's revenue data).[47] We additionally note that:

  (i) UpCloud provided data based on credit usage, rather than revenue, which gives a slightly different result and noted that services were charged in USD between 2018–2022 and in EUR from 2023 onwards.[48]

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[39] AWS' responses to the CMA's information requests [✂].
[40] Microsoft's response to the CMA's information request [✂].
[41] Google's response to the CMA's information request [✂].
[42] While the data we requested from Oracle for 2021–2023 did not have a single-year spend threshold for defining a newly acquired customer for each year, we anticipate that the data provided by Oracle based on their definition of newly acquired customer may result in an overestimate of Oracle's UK shares of supply by newly acquired customers and revenues from newly acquired customers, for these years relative to other providers, as additional customers below the $100 threshold may be included in its submission. Oracle's response to the CMA's information request [✂].
[43] Oracle's response to the CMA's information request [✂].
[44] While the data we received from IBM for 2021–2023 did not use a single-year spend threshold for defining a newly acquired customer for each year, we anticipate that the data provided by IBM based on their definition of newly acquired customer would result in an overestimate of IBM's UK shares of supply by newly acquired customers and revenues from newly acquired customers, for these years relative to other providers, as additional customers below the $100 threshold may be included in its submission. IBM's response to the CMA's information request [✂].
[45] IBM's response to the CMA's information request [✂].
[46] Other providers include Centerprise, Civo, CoreWeave, [✂], Hyve, Scaleway, Wasabi and UpCloud.
[47] We note that some smaller providers resubmitted historical data to accord with this definition of a new customer, and an additional provider submitted data following the publication of our PDR.
[48] UpCloud's response to the CMA's information request [✂].

(ii) [✂] provided the total number of customers it believed to be in the UK that were acquired by the end of each calendar year.[49]

D.32 AWS provided data on UK customer flows according to how customers are classified within its customer relationship management system. This data therefore includes [✂].[50] However, it provided data on UK revenues by reference to [✂].[51] AWS said the 2023 figures were based on a different methodology for identifying a 'UK customer' to the 2018 to 2022 figures, [✂].[52] While these definitions are not entirely consistent, they are unlikely to have a material effect on the analysis.

D.33 We tested the sensitivity of the findings to the assumption used to define a new customer by requesting from Microsoft, AWS and Google, data on the number of new customers and revenue from new customers, based on different spend thresholds ($500, $1,000, $10,000).[53] We compared this to the same data based on a threshold of $100 for 2023 and 2024.

D.34 A limitation with this sensitivity analysis was that we could not calculate the actual shares because we did not have the sensitivity results from all other cloud providers. Therefore, we compared the shares just between Microsoft, AWS and Google.

D.35 Our sensitivity analysis for the revenue from new customers showed that the findings were robust to changes in the definition of a new customer.

(a) We found that increasing the threshold at which a new customer is defined increases the revenues from new customers. This is because while some low spend customers are now excluded, some higher spend customers are now included which would not have been counted at the lower threshold as they have already spent above the lower threshold amount.

(b) In 2023 and 2024, Microsoft had a larger share than both Google and AWS in all three other thresholds, though its share dropped as the threshold increased.[54]

Shares of supply by newly acquired UK customers

D.36 For each firm, we divided its total number of new customers it acquired in each year by the sum of all providers' new customers to give the share.

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[49] [✂] response to the CMA's information request [✂].
[50] AWS' response to the CMA's information request [✂].
[51] AWS' response to the CMA's information request [✂].
[52] AWS' responses to the CMA's information requests [✂].
[53] CMA analysis of revenue and customer acquisition sensitivity data from Microsoft, AWS and Google. Responses to the CMA's information requests [✂].
[54] Oracle had a higher share of revenue from new customers than Microsoft in 2024. We consider this to be the result of a one-off event which will only increase its share of revenue from new customers to that extent in 2024. Oracle submitted that the reason for the large increase of revenue in 2024 from new customers is associated with the acquisition of Cerner customer contracts in FY24. Oracle's response to the CMA's information request [✂].

D.37 UK shares of supply by newly acquired customers for 2021–2024 are presented in the table below.

Table D.6: UK shares of supply by newly acquired customers, 2021 – 2024

				%
	2021	*2022*	*2023*	*2024*
Microsoft	[50-60]%	[50-60]%	[50-60]%	[50-60]%
Google	[5-10]%	[5-10]%	[5-10]%	[5-10]%
AWS	[10-20]%	[10-20]%	[5-10]%	[5-10]%
Oracle	[0-5]%	[0-5]%	[0-5]%	[5-10]%
IBM	[0-5]%	[0-5]%	[0-5]%	[0-5]%
CoreWeave	[0-5]%	[0-5]%	[0-5]%	[0-5]%
Other	[10-20]%	[10-20]%	[20-30]%	[20-30]%

Source: CMA analysis of data from AWS, Microsoft, Google, IBM, Oracle, CoreWeave and some smaller IaaS providers. Shares may not sum to 100 due to rounding. 'Other' category includes Centerprise, Civo, [✂], Hyve, Scaleway, Wasabi and UpCloud.

D.38 Our analysis shows that:

(a) Microsoft won more than [50-60]% of new customers in each year from 2021 to 2024.

(b) AWS' share of new customers was approximately [10-20]% in 2021 and 2022, decreasing to around [5-10]% in 2023 and 2024.

(c) Google's share of new customers is consistent with its other shares of supply presented: it won [5-10]% of new customers in 2021 and [5-10]% in 2024.

(d) Oracle's share of new customers was around [0-5]% in from 2021 to 2023. In 2024, Oracle's share was higher, at around [5-10]%.

(e) CoreWeave and IBM's shares of new customers were both [0-5]% in each year from 2021 to 2024.[55]

D.39 Shares of supply by newly acquired customers show that Microsoft is winning customers at a significantly higher rate than other cloud providers. It won more than twice as many customers in each of 2021 to 2023 as AWS, and at least three times as many as Google.

D.40 Our sensitivity analysis for the number of new customers showed that the findings were robust to changes in the definition of a new customer. Specifically:

[55] CMA analysis of revenue and customer acquisition data from Microsoft, AWS, Google, Oracle, CoreWeave and IBM. Responses to the CMA's information requests [✂]. Responses to the CMA's information requests [✂].

(a) In 2023, Microsoft continued to have the largest share of number of new customers in the first two scenarios ($500 and $1,000), though its share dropped as the threshold increases. In the final scenario ($10,000), AWS' share of new customer was the largest. This could suggest AWS has a larger proportion of existing customers who are increasing their spend.

(b) In 2024, Microsoft continued to have the largest share in all three other scenarios, though its share dropped as the threshold increases. In particular, for the final scenario (where the threshold is $10,000), the shares of Microsoft, Google and AWS were all within a range of approximately [5-10%].[56]

[56] CMA analysis of revenue and customer acquisition sensitivity data from Microsoft, AWS and Google. Responses to the CMA's information requests [✂].